INTEREST RATE MODELS – VASICEK, CIR, HJM

STRUCTURES, FUNDATIONS AND PRACTICAL IMPLICATIONS

Quantitative Financial Modelling Project — MSc in Financial Risk and Data Analysis

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INTEREST RATE MODELS (VASICEK, CIR, HJM)

I. Derivation of Vasicek bond price formula

Vasicek introduced in 1977 the first model to capture the mean reversion property of interest rates.

1. Vasicek Short Rate Dynamics

The Vasicek model is defined by:

$$dr_t = b\,(\frac{a}{b} - r_t)\,dt + \sigma dW_t$$

where:

- b is the speed of mean reversion
- $\frac{a}{b}$ is the long-term mean level
- σ is the volatility parameter
- W_t is a standard Brownian motion under the risk-neutral measure.

The solution of SDE is:

$$r_t = r_0 e^{-bt} - \frac{a}{b}(e^{-bt} - 1) + \sigma \int_0^t e^{-b(t-s)} dW_s$$

Or, in the conditional form:

$$r_s = r_t e^{-b(s-t)} - \frac{a}{b}(e^{-b(s-t)} - 1) + \sigma \int_t^s e^{-b(s-u)} dW_u \qquad for\ s \geq t$$

2. Zero-Coupon Bond Price

The price at time *t* of a zero-coupon bond maturing at T is:

$$B(t,T) = \mathbb{E}^Q \left[e^{-\int_t^T r_s ds} \mid \mathcal{F}_t \right]$$

Here, we take the expectation under the risk-neutral measure *Q*, conditioned on the information available at time *t*.

Since the Vasicek short rate process is Gaussian, the integral

$$\int_t^T r_s\, ds$$

is normally distributed. Therefore, the exponential of this integral follows a lognormal distribution.
We can then compute the expectation using the lognormal moment-generating function:

$$\mathbb{E}^Q\left[e^{-\int_t^T r_s ds}\right] = e^{-\mathbb{E}\left[\int_t^T r_s ds\right] + \frac{1}{2}Var\left(\int_t^T r_s ds\right)}$$

3. Expectation of the Integral
From the Vasicek dynamics, the conditional mean of r_s given $\mathcal{F}_t$ is:

$$\mathbb{E}^Q[r_s|\mathcal{F}_t] = r_t e^{-b(s-t)} - \frac{a}{b}(e^{-b(s-t)} - 1)$$

Therefore, the expected value of the integral is:

$$\mathbb{E}\left[\int_t^T r_s\, ds\right] = \int_t^t \mathbb{E}[r_s|\mathcal{F}_t]\, ds = \int_t^T r_t\, e^{-b(s-t)} ds + \ terms\ not\ depending\ on\ r_t$$

After a change of variables, we obtain:

$$r_t\left(\frac{1 - e^{-b(T-t)}}{b}\right) = n(t,T) r_t$$

4. Variance of the Integral
The variance of $\int_t^T r_s\, ds$ is:
Recalling $r_s = r_0 e^{-bs} - \frac{a}{b}(e^{-bs} - 1) + \sigma \int_0^s e^{-b(s-u)}\, dW_u$, only the stochastic term $\sigma \int_0^s e^{-b(s-u)}\, dW_u$ contributes to the variance. Therefore, we can write:

$$Var\left(\int_t^T r_s\, ds\right) = Var\left(\sigma \int_t^T \int_0^s e^{-b(s-u)}\, dW_u\, ds\right)$$

Applying Fubini's theorem:

$$\int_t^T \int_0^s e^{-b(s-u)} dW_u\, ds = \int_0^t \int_t^T e^{-b(s-u)} ds\, dW_u + \int_t^T \int_u^T e^{-b(s-u)} ds\, dW_u$$
$$= \int_0^T \int_{max(t,u)}^T e^{-b(s-u)} ds\, dW_u$$

Let's now define

$$K(u) := \int_{max(t,u)}^T e^{-b(s-u)}\, ds = \begin{cases} \dfrac{1 - e^{-b(T-t)}}{b} & if\ u < t \\ \dfrac{1 - e^{-b(T-u)}}{b} & if\ u \geq t \end{cases}$$

Hence:

$$Var\,(\int_t^T r_s\,ds\,) = Var\,(\,\sigma\int_t^T\int_0^s e^{-b(s-u)}dW_u\,ds\,) = Var\,(\sigma\int_0^T K(u)\,dW_u)$$

By Ito isometry:

$$Var\,(\sigma\int_0^T K(u)\,dW_u) = \sigma^2\int_0^T K(u)^2du$$

Now, the integral $\sigma^2\int_0^T K(u)^2du$ is equivalent, through a change of variable and some computations, to:

$$\sigma^2\int_0^T K(u)^2du = \frac{\sigma^2}{2b^3}(1-e^{-b(T-t)})^2\,(1-e^{2bt})$$

5. Final Formula

Recalling the formula $\mathbb{E}^Q\left[e^{-\int_t^T r_s ds}\right] = e^{-\mathbb{E}\left[\int_t^T r_s ds\right]+\frac{1}{2}Var\left(\int_t^T r_s ds\right)}$, we obtain:

$$B(t,T) = \mathbb{E}^Q\left[e^{-\int_t^T r_s ds}\mid F_t\right] = e^{-\mathbb{E}\left[\int_t^T r_s ds\right]+\frac{1}{2}Var\left(\int_t^T r_s ds\right)}$$
$$= e^{-r_t\left(\frac{1-e^{-b(T-t)}}{b}\right)+\frac{\sigma^2}{4b^3}\left(1-e^{-b(T-t)}\right)^2\left(1-e^{2bt}\right)} = e^{m(t,T)+n(t,T)r_t}$$

Where

$$n(t,T) = -(\frac{1-e^{-b(T-t)}}{b})$$

And

$$m(t,T) = \frac{\sigma^2}{4b^3}(1-e^{-b(T-t)})^2\,(1-e^{2bt})$$

II. Derivation of CIR bond price formula

1. Cox-Ingersoll-Ross (CIR) model

The CIR model, introduced in 1985 as an improvement over the Vasicek model, is one of the affine models used to describe the dynamics of the short-term interest rate r_t. The main difference compared to Vasicek lies in the diffusion component, where the square root of the rate appears, in order to avoid the consideration of negative interest rates' values.

The stochastic differential equation (SDE) that dictates the behavior of the rate into this scenario is the following:

$$dr_t = k(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$$

- k is the speed of mean reversion
- θ is the long-term mean
- σ is the volatility
- W_t is the Standard Brownian Motion.

2. Why the CIR bond price cannot be derived via direct expectation like in Vasicek

In both the Vasicek and CIR models, the price of a zero-coupon bond maturing at time T is given theoretically by the risk-neutral valuation formula:

$$B(t,T) = \mathbb{E}^Q\,[e^{-\int_t^T rs ds} \mid F_t]$$

However, while this expectation can be solved explicitly in the Vasicek model, it cannot be in the CIR model. This difference stems from the distribution of the short rate process:

- In Vasicek, the short rate is a Gaussian process, so $\int_t^T r_s\, ds$ is normally distributed. This makes the expectation analytically tractable using standard moment-generating function techniques.
- In CIR, the short rate follows a Cox-Ingersoll-Ross (square-root) process, which is non-Gaussian and strictly positive (under the Feller condition). As a result, the distribution of the integral $\int_t^T r_s\, ds$ is not standard and does not lead to a closed-form expectation.

Therefore, in the CIR case, we solve a partial differential equation (PDE) derived from the pricing dynamics, assuming an Ansatz (trial) solution of the exponential-affine form:

$$B(t,T) = m(t,T)e^{-n(t,T)rt}$$

This leads to a system of Riccati differential equations for *n(t,T)* and *m(t,T)*, which gives the final closed-form expression.

3. Bond price and main assumptions

According to the above dynamics, the bond price *B(t,T)* with expiration *T* is:

$$B(t,T) = \mathbb{E}_t(e^{-\int_t^T r_s ds})$$

and it satisfies the partial differential equation (PDE)

$$\frac{\partial B(t,T)}{\partial t} + k(\theta - r_t)\frac{\partial B(t,T)}{\partial r} + \frac{1}{2}\sigma^2 r_t \frac{\partial^2 B(t,T)}{r^2} - rB(t,T) = 0$$

Two fundamental assumptions must be added; the PDE must be solved by assuming the 'trial function' or Ansatz solution:

$$B(t,T) = m(t,T)e^{-n(t,T)rt}$$

and that $B(T,T)$ must be equal to 1.

4. Calculation steps

First, we calculate the three required partial derivatives, with respect to time (t) and the short-term interest rate (r_t):

$$\frac{\partial B(t,T)}{\partial t} = \left(\frac{\partial m}{\partial t} - r\frac{\partial n}{\partial t}n\right) e^{-nr} \qquad \frac{\partial B(t,T)}{\partial r} = -nme^{-nr} \qquad \frac{\partial^2 B(t,T)}{r^2} = e^{-nr}mn^2$$

These derivatives are then plugged in the SDE:

$$\left(\frac{\partial m}{\partial t} - r\frac{\partial n}{\partial t}n\right)e^{-nr} + k(\theta - r_t) - nme^{-nr} + \frac{1}{2}\sigma^2 r_t e^{-nr}mn^2 - rme^{-nr} = 0$$

After some mathematical manipulations, we obtain an equality with terms depending on r_t and others not. Splitting in two systems helps to find the entire structure of the CIR bond pricing formula:

$$\begin{cases} \frac{\partial n}{\partial t} = kn + \frac{1}{2}\sigma^2 n^2 - 1 \\ n(T,T) = 0 \end{cases} \quad \begin{cases} \frac{1}{m}\frac{\partial m}{\partial t} = kn\theta \\ m(T,T) = 1 \end{cases}$$

We know that $B(T,\,T) = 1$ and that the Ansatz solution must also hold at maturity, meaning that $m(T,\,T)e^{-n(T,T)rt} = 1$. The only way to obtain 1 is by inserting the two constraints $n(T,T) = 0$ and $m(T,T) = 1$.

The first system can be solved with the Riccati equation, leading to the well-known solution in the CIR model:

$$n(t,T) = \left(\frac{2(e^{(T-t)\gamma}-1)}{2\gamma+(k+\gamma)(e^{(T-t)\gamma}-1)}\right) \qquad \text{where } \gamma = \sqrt{k^2 + 2\sigma^2}$$

After founding $n(t,T)$, the second system can also be solved, providing the other solution component of the formula.

$\frac{1}{m}\frac{\partial m}{\partial t}$ can be also written as $\frac{d}{dt} log\ \ m(t,T)$, and then taking the integral of both sides we will obtain:

$$log\, m(T,T) - log\, m\,(t,T) = \int_t^T k\theta n(s,T)ds$$

$$log\ m\,(t,T) = -k\theta \int_t^T n(s,T)ds$$

The final result for the other system is:

$$m(t,T) = \left(\frac{2\gamma e^{\frac{(k+\gamma)(T-t)}{2}}}{2\gamma+(k+\gamma)\left(e^{(T-t)\gamma}-1\right)}\right)^{\frac{2k\theta}{\sigma^2}} \qquad \text{where } \gamma = \sqrt{k^2+2\sigma^2}$$

III. The Heath-Jarrow-Morton (HJM) Framework and its calibration challenges

Modeling interest rates through the instantaneous short rate has both advantages and limitations. To address some of these limitations, Ho and Lee (1986) proposed an alternative by modeling the evolution of the entire yield curve within a binomial tree framework. Building on this idea, Heath, Jarrow, and Morton (1992) developed a continuous-time framework that models the entire forward rate curve directly.

The instantaneous forward rate in the HJM framework is given by:

$$f(t,s) = R(t,s,T) = -\frac{d\ ln\ B(t,s)}{ds}$$

This result can be derived by considering the definition of the forward return:

$$R(t,T) = -\frac{1}{T-t}\ ln\ B(t,T)$$

By taking instantaneous forward rates as the primary modeling variables, the HJM framework ensures an arbitrage-free evolution of the entire yield curve. In this setting, the dynamics of forward rates are entirely determined by the volatility structure. This contrasts with one-factor short-rate models, where specifying the volatility of the short rate alone is not sufficient to fully define the behavior of interest rates.

In the HJM framework, for each maturity *T*, the instantaneous forward rate *f(t,T)* follows the stochastic differential equation:

$$df(t,T) = \alpha(t,T)dt + \sigma(t,T)dW_t$$

where:

- *α(t,T)* is the drift term and
- *σ(t,T)* is the volatility term

In the multi-factor extension, *σ(t,T)* becomes an n-dimensional vector function, with each component representing a distinct source of risk. The dynamics then become:

$$df(t,T) = \alpha(t,T)dt + \sum_{i=1}^{n} \sigma_i(t,T)dW_i^t$$

Unlike short-rate models, where the short rate is the state variable and the drift can be freely specified, the HJM framework derives the drift term from the volatility function to maintain arbitrage-free pricing. In order to have a risk-neutral measure, the drift should be defined as following:

$$\alpha(t,T) = \sigma(t,T)\int_t^T \sigma(t,s)ds$$

or in the explicit formulation:

$$\alpha(t,T) = \sum_{i=1}^{n} \sigma_i(t,T)\int_t^T \sigma_i(t,s)ds$$

In that way, the integrated dynamics of $f(t,T)$ follows the formula:

$$f(t,T) = f(0,T) + \int_0^t \sigma(u,T)\,(\int_u^T \sigma(u,s)ds)\,du + \int_0^t \sigma(u,t)\;dWu$$

So the instantaneous forward interest rate equals the initial curve observed at time 0, plus the integrated drift and stochastic components. In both single and multi-factor HJM models, the forward rate *f(t,T)* is entirely determined by the volatility structure, which plays a crucial role in forward curve dynamics.

Given these dynamics of the instantaneous forward rate *f(t,T)*, the application of Ito's lemma gives the following dynamics of the zero-coupon bond price *P(t,T)*:

$$dB(t,T) = B(t,T)\,[r(t)\,dt - (\int_t^T \sigma(t,s)\,ds)\,dW(t)]$$

where r(t) is the instantaneous short-term interest rate at time t, that is:

$$r(t) = f(t,t) = f(0,t) + \int_0^t \sigma(u,t)\int_u^t \sigma(u,s)\,ds\;du + \int_0^t \sigma(s,t)\,dW(s)$$

$$= f(0,t) + \sum_{i=1}^{N}\int_0^t \sigma_i(u,t)\int_u^t \sigma_i(u,s)\,ds\;du + \sum_{i=1}^{N}\int_0^t \sigma_i(s,t)\,dW_i(s)$$

From HJM to Hull-White and Vasicek:

A central limitation of the HJM framework is that, in general, the short rate r_t is not Markovian, as its dynamics depends on the entire history of the volatility function $\sigma(t,T)$. In fact, time *t* appears in the stochastic integral both as extreme of integration and inside the integrand function.

However, this issue can be resolved under a key structural condition. The short rate becomes Markovian if the volatility function is separable, that is:

$$\sigma(t,T) \equiv \xi(t)\psi(T)$$

Then, in the one-factor setting *(N=1)*, the short rate r_t can be written as:

$$r_t = A_t + \psi_1(t) \int_0^t \xi_1^2(u) dW_u$$

Where A_t is defined as:

$$A_t \equiv f(0,T) + \psi_1(t) \int_0^t \xi_1^2(u) \int_u^t \psi_1(s) ds\, du$$

$$dr_t = \left[A_t' dt + \psi_1'(t) \frac{r_t - A_t}{\psi_1(t)}\right] dt + \psi_1(t)\xi_1(t) dW_t^1$$

This form allows the stochastic integral driving the short rate to be expressed directly in terms of r_t, removing the path dependence. Under this assumption, the short rate obeys the SDE of the form:

$$dr_t = (a_t + b_t r_t)dt + c_t dWt$$

This is precisely the Hull-White model, which allows closed-form bond pricing. If we further assume that these coefficients are constant in time, the model simplifies to the well-known Vasicek model:

$$dr_t = a(b - r_t)dt + \sigma dW_t$$

Overfitting in HJM

Overfitting occurs when a model is too flexible and starts capturing random noise in the data rather than the true underlying structure. This is a particular risk in the HJM framework with multi-factor volatility structures. Recall that in the multi-factor case, each component of volatility corresponds to a distinct source of randomness; the more factors included, the more flexible the model can fit different term structure dynamics.

However, as in many areas of statistical learning, a perfect fit to historical data does not necessarily lead to accurate forecasts. In fact, an overly complex model may produce unrealistic projections for the future evolution of the yield curve. From three factors onward, the parameter space becomes high-dimensional, increasing the risk of overfitting and making calibration more difficult.

HJM's lack of closed-form solutions

A major practical limitation of the HJM framework is the general lack of closed-form solutions for pricing interest rate forwards. Since HJM directly models the entire forward rate curve and requires the drift term to be derived from the volatility structure via an integral (to ensure arbitrage-free dynamics), the resulting mathematical complexity often precludes analytical pricing formulas.

This contrasts with short-rate models like CIR or Vasicek, where zero-coupon bond prices and certain derivative prices can be expressed in closed form. In HJM, especially in multi-factor settings, pricing typically relies on numerical techniques such as Monte Carlo

simulations. This increases computational demand, complicates calibration, and limits the framework's practicality for real-time pricing and risk management — unless one adopts simplifying assumptions such as separable volatility, as discussed earlier.

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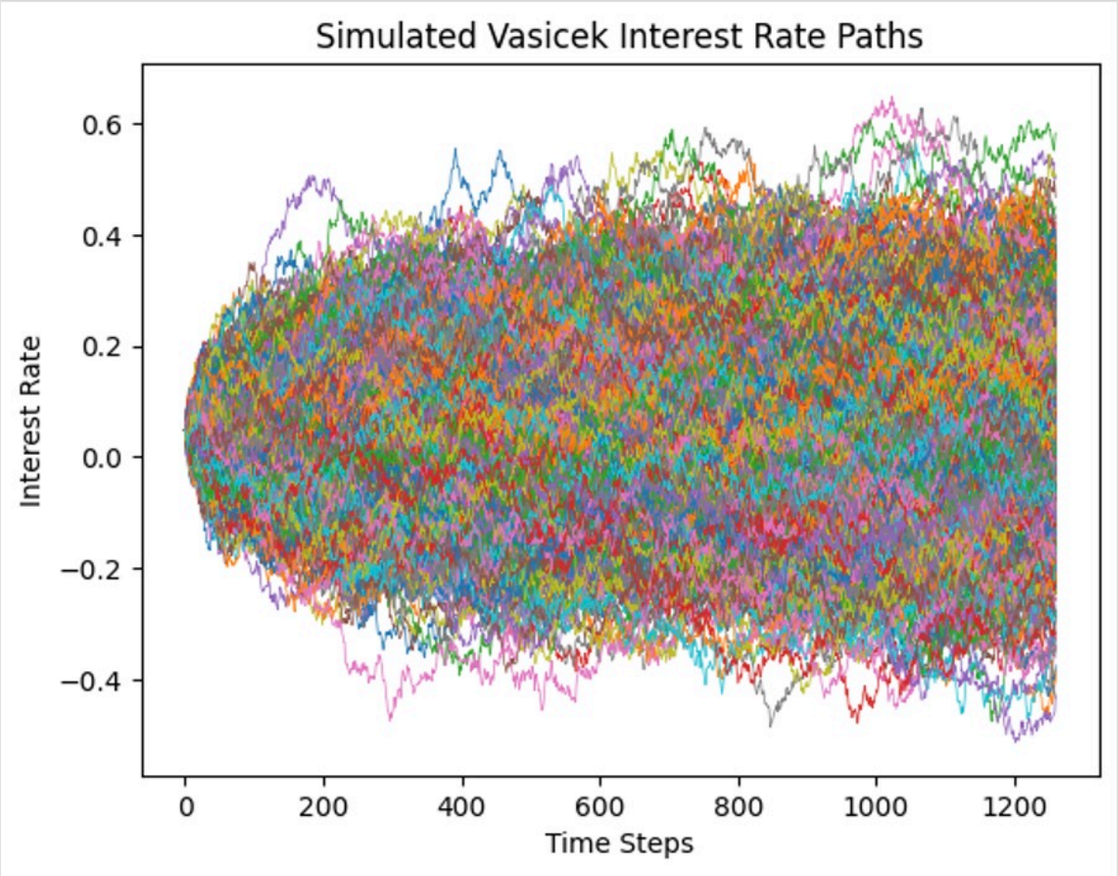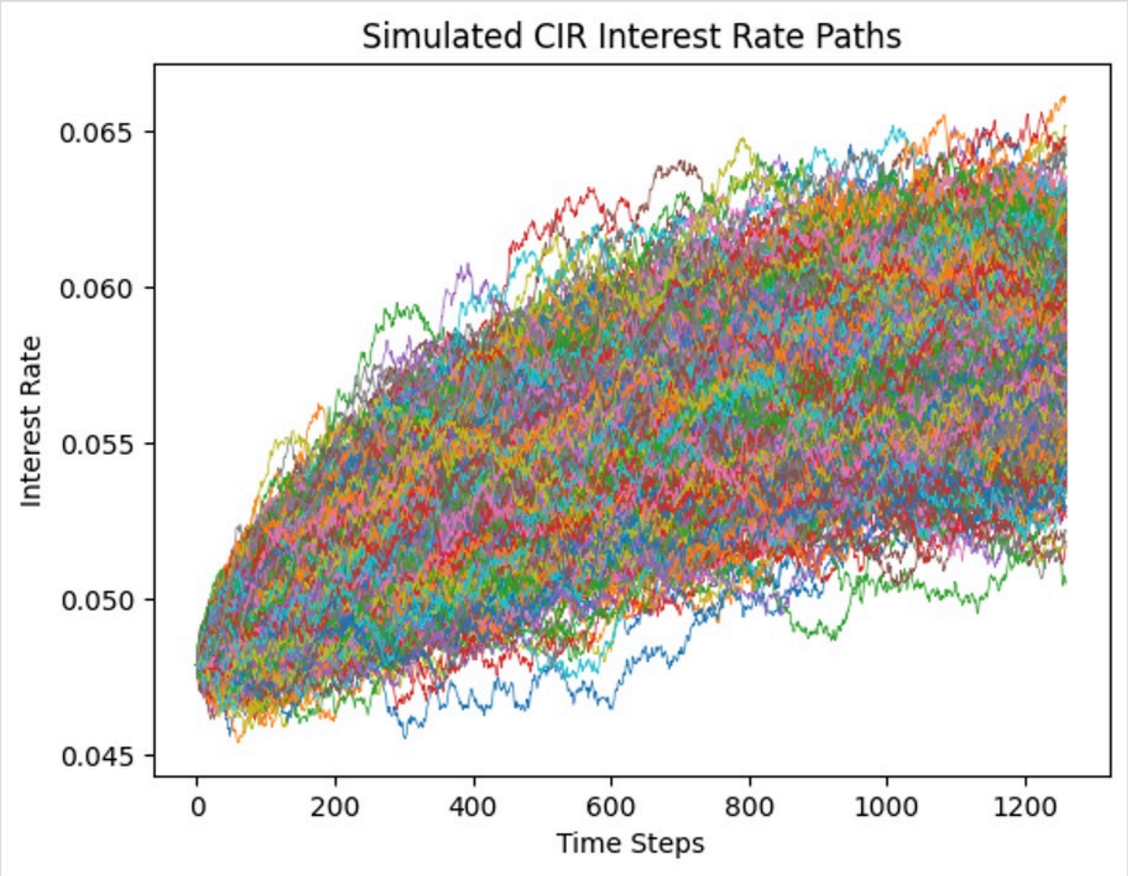
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